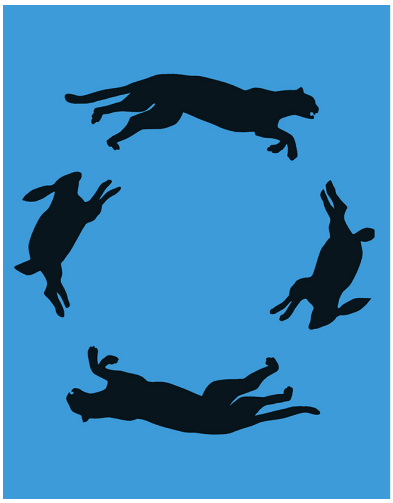


FOOD IS THE BURNING QUESTION

ANIMAL ECOLOGY

IN CONTEXT

KEY FIGURES

Charles Elton (1900–91), **George Evelyn Hutchinson** (1903–91)

BEFORE

Ninth century Arab writer al-Jahiz introduces the idea of the food chain in *Kitab al-Hayawan* (Book of Animals).

1917 American biologist Joseph Grinnell first describes an ecological niche in his paper, “The niche relationships of the California Thrasher.”

AFTER

1960 American ecologist Garrett Hardin publishes an essay in *Science* in which he states that “every instance of apparent coexistence must be accounted for.”

1973 Australian ecologist Robert May’s *Stability and Complexity in Model Ecosystems* uses mathematical modeling to demonstrate that complex ecosystems do not necessarily lead to stability.

2000s Scientists highlight growing concern over ecological tipping points (critical thresholds that, when crossed, cause abrupt shifts between contrasting ecosystems).